



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to VMS FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

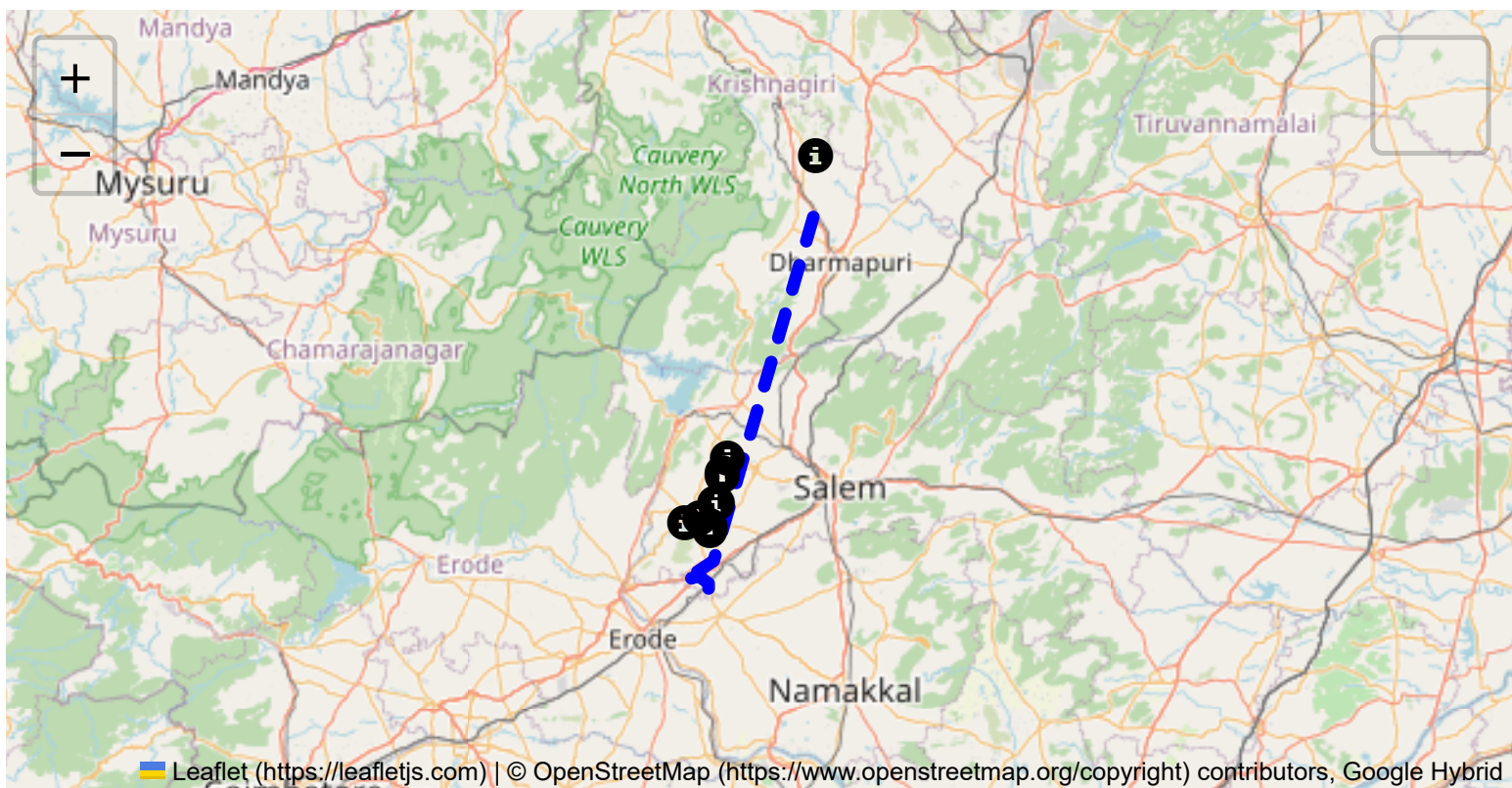
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

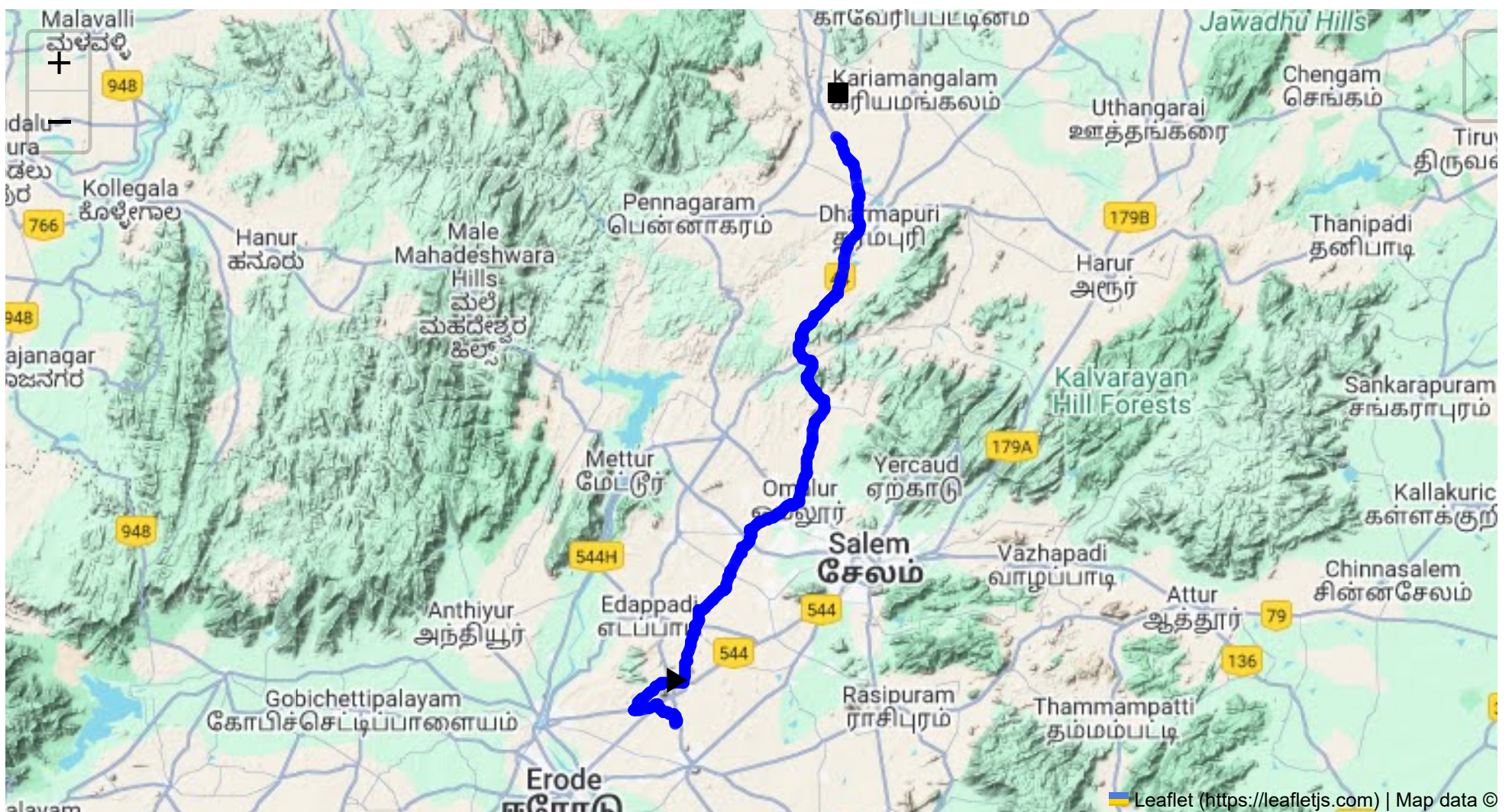
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 116.89 km**  
**Estimated Duration: 2.4 hours**  
**Adjusted Duration (Heavy Vehicle): 3.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.24051, 78.10029)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route covers a significant portion of Tamil Nadu, traversing through various towns and rural areas. Starting from Sangagiri in Tamil Nadu and ending in Somenahalli, the travel path includes key waypoints such as FV2C+2PH Sankari RS - Erode Privu Rd, and several sections along SH-86. The route primarily

follows state highways and major roads, offering a direct path through agricultural and semi-urban regions.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu generally experiences a tropical climate with hot summers, moderate monsoons, and mild winters. Monsoon season, typically from June to September, can bring heavy rainfall.
- **Potential Hazards:** During monsoons, roads can become slippery, and low-lying areas may face flooding, impacting road conditions and visibility. During peak summer months, high temperatures can cause heat-related vehicle issues and discomfort.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Generally, traffic tends to peak in the morning (8:00-10:00 AM) and evening (5:00-8:00 PM) as people commute to and from work.
- **Congestion-prone Areas:** Town centers and market areas in Sankari and Erumaipatti are typically congested. Certain stretches of SH-86 may also see slower traffic, especially near intersections with local roads.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The primary roads such as SH-86 and Bhavani Main Road are generally maintained but may have patches of wear. Areas in rural sections might have narrower roads and less street lighting.
- **Infrastructure:** Infrastructure is adequate for regular vehicular traffic; however, heavy vehicles might face difficulties on narrower village roads or where road maintenance is pending.

## 5. Suggestions for Alternative Routes for Emergencies

In emergencies, SH-86 offers several offshoot routes connecting to NH-44, a major national highway that might provide better access to emergency services and faster routes out of congested areas.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to specific regulations such as obtaining necessary permits and displaying proper signage. Restrictions may apply during certain hours in urban areas to minimize risk and congestion.

## 7. Overview of Historical Incidents

While no specific incidents may have been documented publicly, local reports suggest common risks include vehicle breakdowns and accidents in congested areas. Hazardous material spills, if any, would likely have stricter historical documentation due to regulatory oversight.

## 8. Environmental Considerations and Sensitive Areas

- Water Bodies:** Be cautious near river or canal crossings, especially during monsoon seasons when water levels rise.
- Wildlife Areas:** Some stretches may be near farming zones, approaching wildlife cautiously as stray animals can interfere with traffic.

## 9. Analysis of Communication Coverage

The route enjoys relatively good mobile coverage, but rural patches, especially near agricultural lands, might have spotty signal strength leading to potential communication dead zones.

## 10. Estimated Emergency Response Times

- Urban Areas:** Response times are typically quicker in and around towns such as Sangagiri and Erumaipatti (within 30 minutes).
- Rural Sections:** More isolated areas may experience longer response times, potentially exceeding 45 minutes to an hour. It is advisable to clarify procedures with local service providers.

## 12. Overall Summary of Risk Assessment

Overall, the route is moderately safe with careful attention needed during peak traffic hours, poor weather conditions, and in rural segments. Proper planning, compliance with local regulations, and communication readiness can mitigate most risks. It is advisable to frequently monitor weather forecasts, maintain efficient communication equipment, and ensure drivers are well-versed with emergency procedures.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2 | Turn      | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 3 | Turn      | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 5 | Turn      | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 6 | Turn      | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 7 | Turn      | Medium     | 11.45571, 77.81702 | 30 KM/Hr    |
| 8 | Turn      | High       | 11.45615, 77.81677 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 9  | Blind Spot | Blind Spot | 11.49395, 77.88537 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49465, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.56096, 77.89931 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.57035, 77.90223 | 30 KM/Hr    |
| 13 | Turn       | High       | 11.57041, 77.90221 | 15 KM/Hr    |
| 14 | Turn       | Medium     | 11.57649, 77.90610 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.58279, 77.90658 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.63833, 77.94818 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.64171, 77.95184 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.64243, 77.95198 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.64721, 77.95118 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.67870, 77.96846 | 30 KM/Hr    |
| 21 | Turn       | High       | 11.70227, 77.97864 | 15 KM/Hr    |
| 22 | Turn       | Medium     | 11.71715, 77.99969 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.71921, 78.01133 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.73775, 78.03465 | 30 KM/Hr    |
| 25 | Turn       | High       | 11.73651, 78.03878 | 15 KM/Hr    |
| 26 | Turn       | Medium     | 11.74309, 78.04725 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.86873, 78.08374 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.87966, 78.08187 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.92609, 78.06513 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.93719, 78.05372 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.95457, 78.04685 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.95828, 78.04983 | 15 KM/Hr    |
| 33 | Turn       | Medium     | 11.95885, 78.04988 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.96044, 78.04949 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.96335, 78.05141 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.96408, 78.05131 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 11.96511, 78.04966 | 30 KM/Hr    |
| 38 | Turn       | High       | 11.96608, 78.04919 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 11.96819, 78.05088 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 12.12075, 78.13278 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 1  | hospital | Sri Shanthi Hospital               | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2  | hospital | Sugumar Hospital                   | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3  | hospital | Omalur Government Hospital         | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 4  | clinic   | R.R. Nursing Home                  | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 5  | hospital | Sri Hari Hospital                  | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 6  | hospital | Sri Sugam Hospital                 | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 7  | hospital | Saravana Hospital                  | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8  | hospital | Arun Hospital                      | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 9  | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 11 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 12 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 13 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 14 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 15 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 16 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 17 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 18 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 19 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 20 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 21 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 23 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 25 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 27 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 28 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 29 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |

## Crowded Spots

|    | type    | name                                                 | coordinates               | speed_limit | risk_level |
|----|---------|------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                           | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 10 | school  | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 24 | school  | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 26 | college | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |
| 30 | college | Laxminarayana Polytechnic College                    | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45898, 77.85135



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45571, 77.81702



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45615, 77.81677



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49395, 77.88537



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49465, 77.88488



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56096, 77.89931



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57035, 77.90223



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.57041, 77.90221



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57649, 77.90610



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58279, 77.90658



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.63833, 77.94818



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64171, 77.95184



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64243, 77.95198



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64721, 77.95118



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.67870, 77.96846



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70227, 77.97864



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71715, 77.99969



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71921, 78.01133



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73775, 78.03465



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73651, 78.03878



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.74309, 78.04725



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.86873, 78.08374



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.87966, 78.08187



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.92609, 78.06513



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.93719, 78.05372

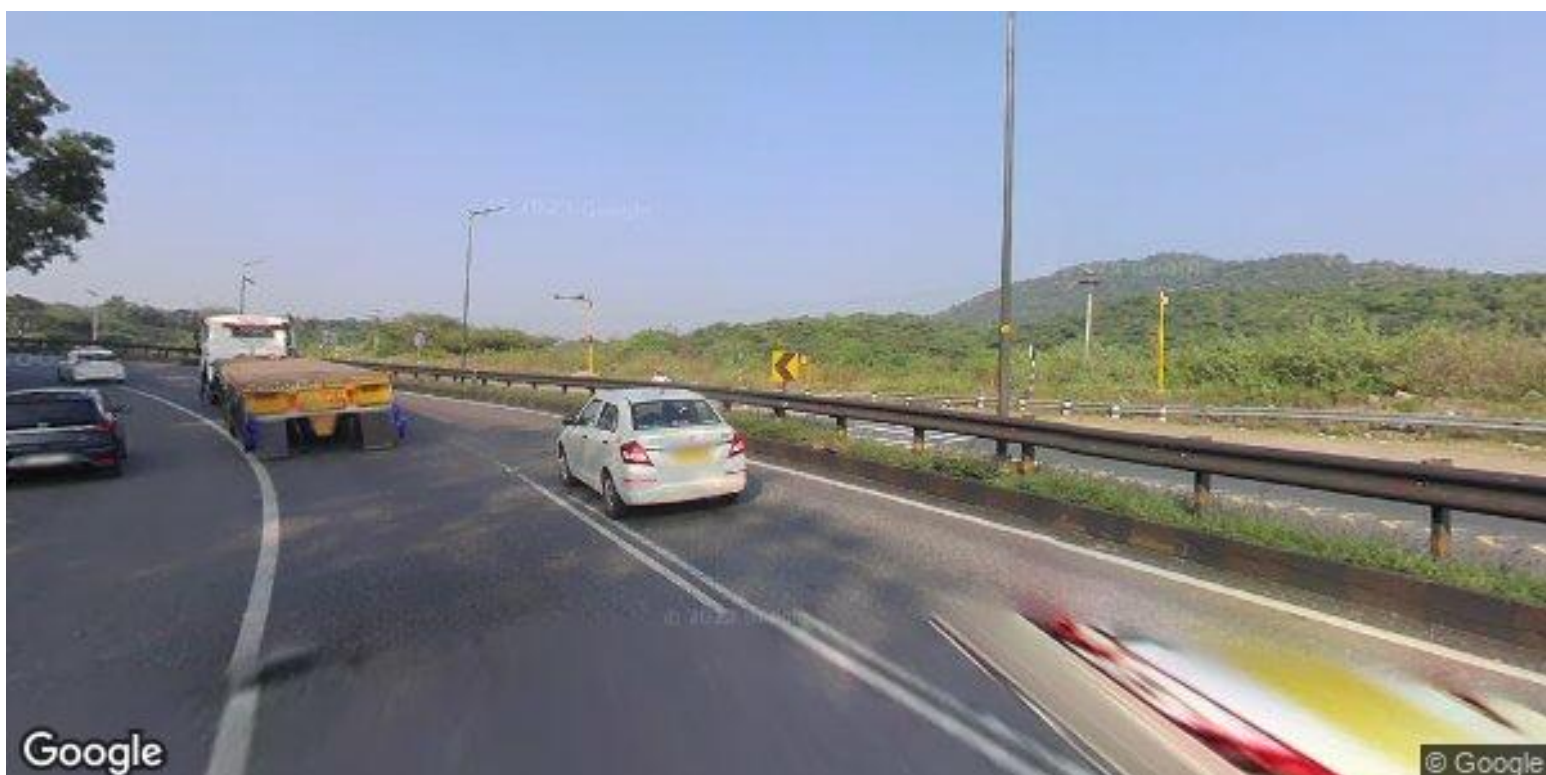


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95457, 78.04685



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95828, 78.04983



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95885, 78.04988



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96044, 78.04949



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96335, 78.05141



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96408, 78.05131



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96511, 78.04966



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96608, 78.04919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96819, 78.05088



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.12075, 78.13278

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